Game Theory-Based Decision-Making and Iterative Predictive Lateral Control for Cooperative Obstacle Avoidance of Guided Vehicle Platoon

Xinle Gong, Sheng Liang, Bowen Wang, and Wei Zhang

***Abstract*—This paper presents a systematic decision-making and lateral control framework to realize cooperative obstacle avoidance (OA) of the guided vehicle platoons safely and integrally in multiple scenarios. In the control framework, a centralized decision-making strategy is developed to determine optimal overtaking maneuver (i.e., OA) in the dynamic driving environment, and a distributed lateral controller with a safe driving corridor planner is designed for lateral reference trajectories tracking. The proposed decision-making strategy formulates the influence mechanism between the controlled platoon and potential surrounding vehicles based on an interactive behavior model. Moreover, a game theory-based algorithm is applied to calculate safe timing of overtaking and select efficient overtaking mode while avoiding the negative impact on the surrounding vehicles. For the distributed lateral control strategy, the control problem is solved by a linear time-varying model predictive control (LTV-MPC) algorithm so as to reduce the computational burden. Furthermore, to improve the accuracy of LTV-MPC algorithm in varying scenarios, a tailored iterative control logic is presented. The performance of the proposed framework is evaluated on the dSPACE hardware-in-loop platform. The results show that the proposed framework works well under various test scenarios involving multiple static and moving obstacles.**

***Index Terms*—Guided vehicle platoon, lateral control, obstacle avoidance (OA), decision-making, iterative linear time-varying model predictive control (ILTV-MPC).**

I. Introduction

A. Background and Motivation

DUE to the high complexity of modern traffic system and increase of vehicle population, the vehicle crashes caused by human errors are on the rise, which poses a serious challenge to traffic safety and efficiency in recent decades. With the rapid development of on-board sensors, computation, and communication technologies, cooperative control technology of multiple autonomous vehicles is believed to be a possible solution to improve traffic safety and efficiency and becomes one of the leading research areas. [1], [2], [3].

To enhance the transportation efficiency and reduce the cost of human drivers and on-board sensors, a variety of researches on the guided intelligent and connected vehicle platoon are conducted in recent years. Generally, a guided vehicle platoon consists of a leading vehicle and several followers (commonly 2–5 followers), only the leading vehicle equips with high performance sensors and computing platform for environment perception. Therefore, a cooperative control strategy for the guided platoon should be utilized to keep the platoon intact while travelling. Of current platooning control technologies, the majority focus on longitudinal control design, such as adaptive cruise control (ACC) and cooperative ACC (CACC) [4], [5], [6]. However, to enable autonomous vehicles to maintain in a platoon fashion, it is necessary to realize coordination control of the longitudinal and lateral motions because frequent vehicle operations (e.g., obstacle avoidance (OA)) need to be accomplished by steering on daily road circumstance.

B. Related Works

The development of lateral control strategy for the vehicle platoon is a daunting task that entails the planning and tracking of an optimal trajectory while considering complex road and traffic environment constraints (e.g., limited free space available for overtaking). An intuitive way to solve the lateral control problem of the vehicle platoon is the vehicle following mode, where each vehicle tries to maintain the platooning formation by only following its direct preceding vehicle [7], [8]. The advantage of this approach is that the relative lateral position between the following and the preceding vehicle can be directly measured and employed as the lateral reference trajectory after proper treatment. However, the vehicle following strategy could result in corner-cutting. String stability [9], [10], [11] also needs to be considered when projecting a controller for such structure. An early definition of the lateral string stability for the vehicle platoon can be seen in [12], where three lateral control methods for autonomous vehicle following are discussed and the importance of inter-vehicle communication to the lateral platoon stability is demonstrated. An integrated platoon control framework for heterogeneous vehicles is proposed in [13], in which the followers track the trajectory of the leading vehicle instead of its direct preceding vehicle. The string stability of

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the platoon is ensured by a H_∞ platoon controller based on the Lyapunov-Krasovskii functional.

Another popular approach to implement the lateral control of the vehicle platoon is lane following where all vehicles within the platoon track the same reference trajectory such as the road centerline [14], [15]. Early researches on lane following strategy are concentrated on leverage look-down sensing, where the reference road centerline is obtained by detecting the signal (i.e., magnetic signal) on a road surface by on-board sensors [16], [17], [18], [19]. The application of this method is limited by expensive infrastructure investment. Instead, a look-ahead sensing based approach is developed for the lane following strategy in which forward-looking sensors (e.g., cameras) are utilized to detect lane marking on the road [20]. However, this approach does not work well for the vehicle platoon with a small inter-vehicle distance because the lane making will substantially be obscured by their preceding vehicles [21], [22]. A lateral control method with on-board sensing and inter-vehicle communication is developed in [22], where the lateral displacements of the followers to the road centerline are estimated based on their relative positions to the preceding vehicles and the road curvature received from the leading vehicle. The robustness of the estimation against sensor noise and communication delays are investigated by a Monte-Carlo simulation.

With the development of autonomous driving technologies, more control technologies have been applied to the lateral control of the vehicle platoon. For example, Kianfar et al. [23] designed a distributed model predictive control (DMPC) approach, which guarantees the string stability of the vehicle platoon by enforcing corresponding constraints into the optimization problem. Solyom et al. [24] developed a control framework based on the traditional PID approach to achieve the lateral string stability in the sense of $\mathcal{L}_2$. A sliding-mode based lateral control strategy is introduced in [22], where the control robustness and string stability in the presence of actuation and sensor delays are investigated. Additionally, to address modelling uncertainties and external disturbances, robust control methods are widely studied in vehicle control system, such as H_∞ controller [25], LMI technique [26], [27], and tube-based MPC [28]. For example, in [25], a robust weighted gain-scheduling H_∞ lateral motion controller is designed to minimize the effects of uncertain time-varying longitudinal velocity and tire cornering stiffness. The overall tracking performance is tuned by a weighting factor. On the other hand, parameter estimation is another effective way to handle the modelling errors and external disturbances [29], [30].

Recently, the integrated decision and control (IDC) framework is proposed and applied in some researches [31], [32], [33]. In the IDC framework, several candidate reference paths are firstly generated by a static path planner, then the IDC controller selects and tracks the optimal reference path simultaneously while considering the dynamic obstacles. Simulations and real road tests demonstrate that the IDC can improve the control performance significantly in some specific scenarios, such as the multi-lane intersections with mixed traffic flow.

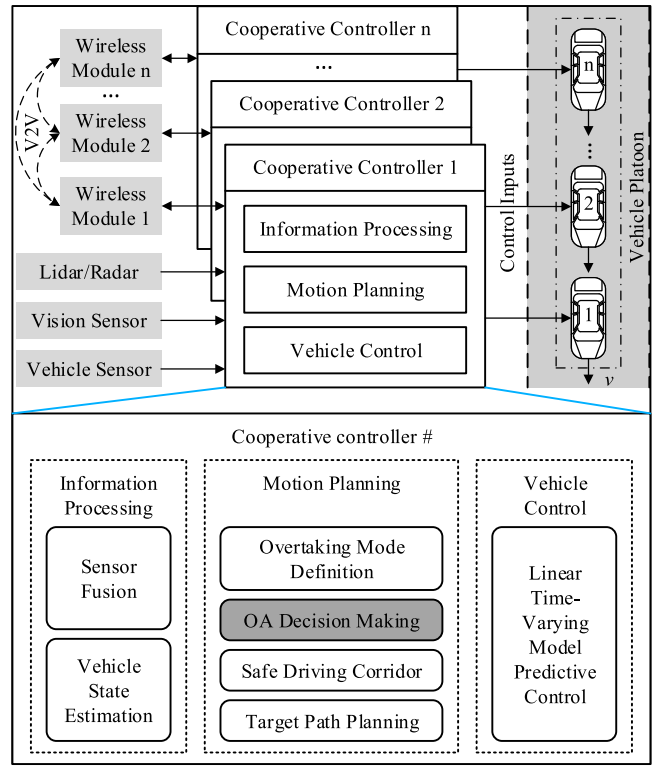

Fig. 1. Framework of the proposed system for the guided vehicle platoon. Light gray blocks denote information sources, such as sensors and wireless communication systems. Dark gray block is a centralized block that is implemented on the leading vehicle.

C. Contributions

For the guided vehicle platoon, a central challenge of lateral control strategies is performing cooperative OA and keeping the platoon intact in a complex driving scenario, especially when the platoon interacts with environmental vehicles in overtaking maneuvers. In this paper, a decision-making and lateral control framework for the guided vehicle platoon is proposed (see Fig. 1), which can achieve dynamic and static obstacle avoidance on the curved road. The main contributions of this article are as follows:

1) Unlike the existing works that achieve an overtaking maneuver with a fixed pattern [34], [35], we propose a centralized decision-making logic to select the optimal OA mode based on overtaking modes analysis of multiple typical scenarios, where a game theoretic algorithm with an interpretable interactive behavior model is developed to determine the optimal overtaking time instant and velocity sequence. The proposed method can finish OA effectively, and the impact on surrounding environment vehicles is significantly reduced.
2) A distributed predictive control algorithm is proposed for lateral motion control of vehicles. To trade off the accuracy and the computational efficiency, the safe driving corridor constraint is formulated as the time-varying piecewise linear function based on the velocity-dependent dynamic map, and a tailored iterative linear time-varying model

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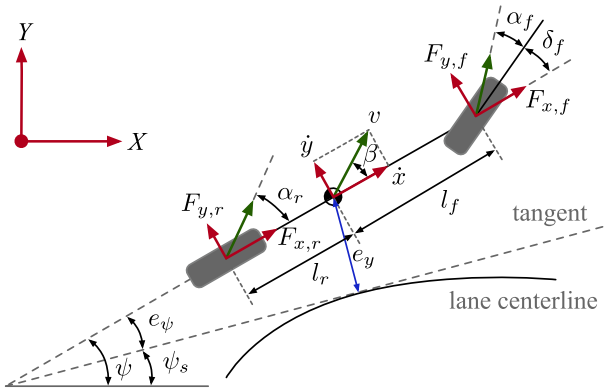

Fig. 2. Nonlinear bicycle model in the Frenet coordinate frame.

TABLE I
PHYSICAL PARAMETERS FOR VEHICLE MODEL

Symbol	Description	Value [units]
m	Vehicle mass	1270 [kg]
I_z	Yaw moment of inertia	1536.7 [kg·m^2]
l_f	Distance from CoG to front axle	1.015 [m]
l_r	Distance from CoG to rear axle	1.895 [m]
$C_{\alpha,f}$	Front wheels cornering stiffness	57300 [N/deg]
$C_{\alpha,r}$	Rear wheels cornering stiffness	86900 [N/deg]
B_f, B_r	Front and rear wheels stiffness factor	0.212, 0.277 [-]
C_f, C_r	Front and rear wheels shape factor	1.300, 1.300 [-]
D_f, D_r	Front and rear wheels peak factor	3734, 2090 [-]
E_f, E_r	Front and rear wheels curvature factor	-0.728, -0.061 [-]

predictive control (ILTV-MPC) algorithm is developed to solve the tracking control problem.

3) The proposed algorithm is implemented in the Hardware-in-loop (HIL) platform based on dSPACE real-time simulator, and experiments under various scenarios are carried out to evaluate the overall performance.

The paper is organized as follows: Section II introduces an overall control architecture of the proposed vehicle platooning system. Modeling details are described in Section III. In Section IV, the specific control laws are presented. Section V analyzes the results of simulation tests. Conclusions are summarized and future works are stated in Section VI.

II. MODELLING

This section introduces the vehicle model that is used as the basis for the control design in the following section. First, a nonlinear vehicle model in the Frenet coordinate frame is presented, and then a discrete LTV representation of the model is derived. It is worth noting that this paper focuses on the development of the lateral control strategy for the vehicle platooning system. Hence, we assume that a longitudinal reference velocity and a desired inter-vehicle distance of the platooning formation can independently be implemented by specific distributed longitudinal motion controllers.

A. Linear Time-Varying Lateral Vehicle Dynamics Model

A nonlinear dynamic bicycle model regarding left and right wheels on each drive shaft as one wheel is employed to describe the lateral motion of the vehicle. The model notations are illustrated in Fig. 2 and the corresponding physical parameters are defined and quantified in Table I.

Using Newton's law and the moment balance, the differential equations are given by

$$\ddot{y} = -\dot{x}\dot{\psi} + \frac{2}{m}(F_{y,f} + F_{y,r}) \tag{1}$$

$$\ddot{\psi} = \frac{2}{I_z}(l_f F_{y,f} - l_r F_{y,r}) \tag{2}$$

where $\dot{x}$ and $\dot{y}$ respectively represent the longitudinal and lateral velocities of the vehicle, in the vehicle body coordinate frame, and $\dot{\psi}$ denotes the yaw rate where ψ is the heading angle with respect to the inertial coordinate frame (X, Y) in Fig. 2. The vehicle mass and yaw inertia are indicated by m and I_z, respectively, and l_f and l_r denote the distances from the front and rear axles to the center of gravity (CoG) of the vehicle, respectively. $F_{y,f}$ and $F_{y,r}$ denote the front and rear lateral tire forces in the vehicle body coordinate frame, respectively, which are calculated approximately by using mathematical tire models, including *Pacejka tire model* and *linear tire model* as follows:

$$F_{y,*} = \begin{cases} -C_{\alpha,*}\alpha_*, & \text{if } |\alpha_*| < \alpha_l \\ -D_* \sin(C_* \tan^{-1}(B_*\alpha_* & \\ -E_*(B_*\alpha_* - \tan^{-1}(B_*\alpha_*)))), & \text{if } |\alpha_*| \geq \alpha_l \end{cases} \tag{3}$$

where $* \in \{f, r\}$, α_* denotes the tire slip angle, and $C_{\alpha,*}$ denotes the tire cornering stiffness. α_l is the critical tire slip angle for model switching and B_*, C_*, D_*, and E_* are constant coefficients that are determined through the experimental data. Utilizing small angle approximations, the tire slip angles can be calculated by

$$\alpha_f = -\delta_f + \frac{\dot{y} + l_f\dot{\psi}}{\dot{x}}, \quad \alpha_r = \frac{\dot{y} - l_r\dot{\psi}}{\dot{x}} \tag{4}$$

where δ_f denotes the steering angle of the front wheel as depicted in Fig. 2.

The dynamic model above is not well-defined at low speeds. As shown in (4), there exists a singularity for the expression of the tire slip angle when the vehicle speed is close to zero. To address this issue, a kinematic bicycle model is employed instead of the dynamic model for the control design at a low speed (10 km/h). The kinematic bicycle model is formulated as follows:

$$\dot{y} = v \sin(\beta) \tag{5}$$

$$\dot{\psi} = \frac{l_r}{v} \sin(\beta) \tag{6}$$

$$\beta = \tan^{-1}\left(\frac{l_r}{l_f + l_r} \tan(\delta_f)\right) \tag{7}$$

where v and β denote the velocity and slip angle of the vehicle's CoG, respectively.

It is of great significance for the application of LK and OA to model the lateral position and orientation of the vehicle relative

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to the Frenet frame [36]. Firstly, the lateral reference position is set as a constant to execute the task of lane changing. Also, obstacle-free road safety constraints are easily described with upper and lower bounds. According to the road-aligned frame as shown in Fig. 2, the following nonlinear equations are used to represent the lateral position and orientation error dynamics,

$$\dot{e_y} = \dot{x}\sin(e_\psi) + \dot{y}\cos(e_\psi) \tag{8}$$

$$\dot{e_\psi} = \dot{\psi} - \frac{\kappa_r}{1-\kappa_r e_y}(\dot{x}\cos(e_\psi) - \dot{y}\sin(e_\psi)) \tag{9}$$

where e_y denotes the distance from the vehicle's CoG to the lane centerline, e_ψ represents the heading angle error between the vehicle and the lane centerline, and κ_r denotes the road curvature.

By combining the lateral vehicle dynamics with the lateral position error dynamics, the nonlinear bicycle model in the Frenet frame can be compactly expressed as

$$\dot{\xi^p}(t) = f^p(\xi^p(t),\, u^p(t),\, p^p(t)) \tag{10}$$

where the superscript p represents an arbitrary vehicle in the platoon. ξ^p is the state vector, $u^p = \delta_f$ is the system control input, and the variable $p^p = [\,\dot{x}, \kappa_r\,]^T$ denotes the parameter vector. The definition of ξ^p depends on the longitudinal velocity of vehicle: if the longitudinal velocity is higher than 10 km/h, the dynamic model (1)–(2) is used, $\xi^p = [\,\dot{y}, \dot{\psi}, e_y, e_\psi\,]^T$; otherwise, the kinematic model (5)–(7) is used, $\xi^p = [\,y, \psi, e_y, e_\psi\,]^T$. Similarly, the definition of f^p also depends on the longitudinal velocity.

The nonlinearities of the vehicle model significantly increase the complexity in real-timely solving the optimization problem for the model predictive controller. To deal with this issue, we linearize the nonlinear bicycle model using a first-order approximation of Taylor expansion around nominal state $\bar{\xi}$ and input $\bar{u}$, and apply the forward Euler approach to discretize the linearized model, thus obtaining an LTV state-space expression in the form of discrete-time as follows

$$\xi^\delta_{k+1} = A_k \xi^\delta_k + B_k u^\delta_k \tag{11}$$

where

$$\xi^\delta_k = \xi_k - \bar{\xi}_k, \quad u^\delta_k = u_k - \bar{u}_k$$

$$A_k = I + \nabla_{\xi_k} f(\xi_k, u_k, p_k)|_{(\bar{\xi}_k, \bar{u}_k, p_k)} \cdot T_s$$

$$B_k = \nabla_{u_k} f(\xi_k, u_k, p_k)|_{(\bar{\xi}_k, \bar{u}_k, p_k)}$$

T_s denotes the sampling time and the superscript p is elided for simplification of the notations.

Remark 1: It is worth noting that since A_k and B_k represent functions of the nominal states and inputs, respectively, their analytic representations can be computed offline for the particular vehicle model engaged, which significantly reduces the real-time computational burden compare to any numerical linearization.

B. Communication Topology and Environment Representation

The communication topology of the controlled guided platoon is shown in Fig. 3. The controlled vehicles in the guided platoon are connected by the vehicle to vehicle (V2V) communication. Specifically, two types of information are assumed to be available for the control system. One is the controlled vehicle i's motion states observed by onboard sensors, like position, direction, velocity, acceleration and lane markings, and the other is motion states of vehicles surrounding the vehicle platoon which are both collected via onboard sensors as well as transmitted from other controlled vehicles' sensors through wireless communication systems. Additionally, vision sensor mounted on the leading vehicle is also able to detect road geometry information, i.e., the curvature and heading angle of the road.

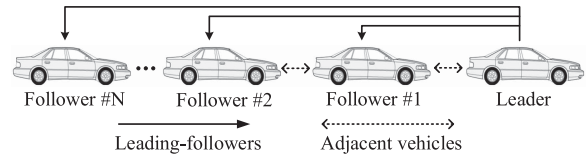

Fig. 3. Communication topology of the guided platoon.

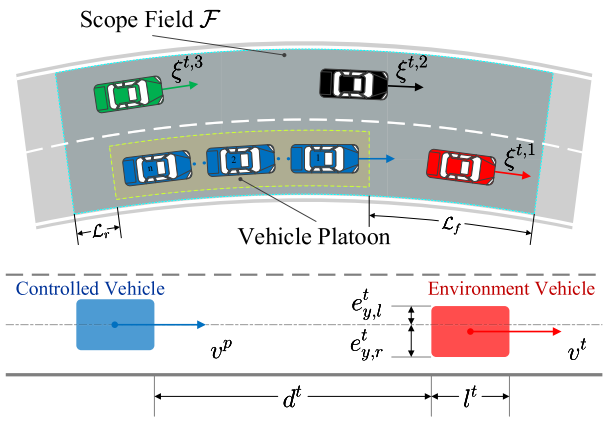

Fig. 4. Environment perception model. Top: illustration of the vehicle platoon and surrounding vehicles in the scope field $\mathcal{F}$ (gray area). Bottom: depiction of measurable states for a single environment vehicle. The environment vehicles are denoted by the superscript t.

Considering the measuring range of the perception system, we define an area near the motorcade along the direction of the road center line as the perception domain $\mathcal{F}$, whose width covers two lanes and length covers the $\mathcal{L}_f$ in front of the first vehicle to the $\mathcal{L}_r$ behind the last vehicle. The notations are illustrated at the top of Fig. 4. In the perception domain, the information of environment vehicles and obstacles can be obtained and shared by all vehicles in the platoon

III. Lateral Control for Cooperative OA

In this section, we focus on design details of the cooperative control strategy for the guided vehicle platoon, which consists of a centralized game theory-based decision-making algorithm implemented on leading vehicle and a distributed predictive lateral controller implemented on each controlled vehicle.

A. Overtaking Mode Analysis

In general, there exist two overtaking maneuvers for a vehicle platoon, the "uniformity" pattern and "snake" pattern. The

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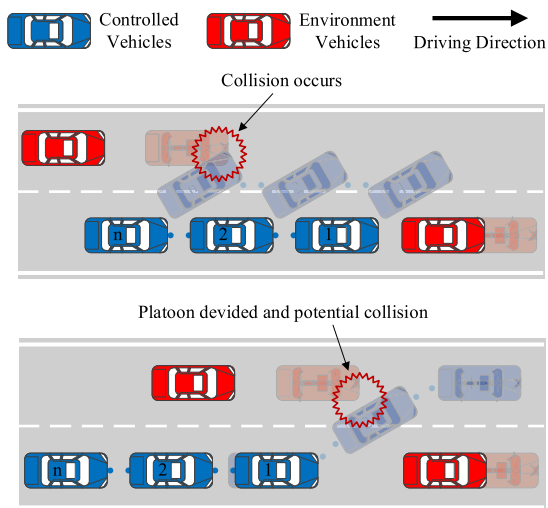

Fig. 5. The potential vehicle collision of two unitary overtaking modes.

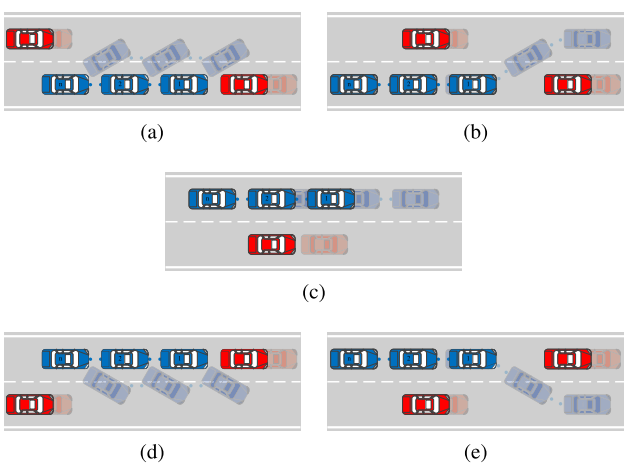

Fig. 6. Three-step overtaking maneuver. (a) and (b) represent the first stage of overtaking. (c) represents the second stage of overtaking. (d) and (e) represent the third stage of overtaking.

"uniformity" mode represents that all vehicles in the platoon change lanes at the same time during the overtaking maneuver and the "snake" mode indicates that each vehicle makes a lane change separately. As stated in the introduction, in terms of the vehicle platoon, the unitary overtaking mode can hardly cope with varying driving scenarios, such as confronting multiple static, dynamically moving or mixed obstacles. Vehicle collision may occur in some situations under both unitary overtaking modes, see Fig. 5. Therefore, different overtaking modes should be defined to adapt to these complex situations. According to the three-step overtaking process: (i) lane-changing to the overtaking lane; (ii) lane keeping (LK); and (iii) returning to the original lane, we design two lane-changing patterns for the guided vehicle platoon at both step 1 and 3 of the overtaking process. Consequently, there mathematically exist four mode combinations available for the implementation of an overtaking maneuver. We denote these mode combinations by $S_o \in \{1, 2, 3, 4\}$, where 1 represents the overtaking mode consisting of a sequential mode combination shown in Fig. 6(b), (c), and (e); 2 for a sequential mode combination in Fig. 6(a), (c), and (d); 3 for a sequential mode combination in Fig. 6(b), (c), and (d); 4 for a sequential mode combination in Fig. 6(a), (c), and (e). An optimal overtaking maneuver will be determined from above four mode combinations by a decision-making algorithm in the following section, provided an OA mode is activated.

B. Decision Making for Obstacle Avoidance

In complex driving circumstances, the vehicle platoon needs to determine the optimal overtaking decision to avoid collision and minimize the impact on surrounding environment vehicles while performing overtaking maneuver. Therefore, we develop a centralized game theoretic decision-making algorithm to coordinate the overtaking modes of the multi-vehicle platoon so as to perform the overtaking maneuver safely and timely. The main objective of the decision-making algorithm is to determine an appropriate lateral driving mode, i.e., LK or OA, for the platooning formation according to the current traffic flow.

To evaluate the control effect and avoid negative impact on environment vehicles during the overtaking process, the interpretable interactive behaviors between the controlled platoon and potential influenced environment vehicles are embodied in the behavior model, and an game theoretic problem with an interactive cost function is used to assess feasibility of overtaking maneuver. In particular, let us denote the sets of the controlled vehicles and the influenced followers by N^p and N^r, respectively. We employ a kinematically longitudinal point-mass model, to build and predict longitudinal motions of the controlled vehicles and the influenced followers, and define their state vector x and input vector u in a different way. For any controlled vehicle $i \in N^p$, we set $x^i = s^i$ and $u^i = v^i$ where s^i is the longitudinal position of the vehicle. For any influenced environment vehicle $j \in N^r$, we set $x^j = [\, s^j, v^j \,]^T$ and $u^j = a^j$ where a^j is the longitudinal acceleration of the vehicle. The state cost of the game theoretic problem at time point t is stated by interactive behavior model as follows:

$$q(x_t, u_t) = \sum_{i \in N^p} q_p(s_t^i, v_t^i, v_{t-1}^i, \tau) + \sum_{j \in N^r} q_r(s_t^j, v_t^j, \tau) \quad (12)$$

where the first term denotes the traveling cost of all controlled vehicles. The second accumulative term indicates the additional cost that the influenced followers need to pay due to the lane-changing behavior of the vehicle platoon.

To reduce the computational load in solving the optimization problem, we here consider the platooning system as a whole where all vehicles are assumed to be controlled synchronously with the same reference velocity. Hence, the state cost function (12) is approximately rewritten as

$$q(x_t, u_t) = N^p q_p(s_t^p, v_t^p, \tau) + \sum_{j \in N^r} q_r(s_t^j, v_t^j, a_t^j) \quad (13)$$

where s_t^p and v_t^p denote the position and average velocity of the vehicle platoon, respectively.

The design of the traveling cost functions q_p and q_r in (13) should be oriented around the goal of improving traffic efficiency while avoiding collisions between vehicles during performing

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Algorithm 1: Decision-Making Logic for OA.

Input: the velocity and position information of all detected objects (environment vehicles) in the scope field $\mathcal{F}$.
Divide objects on the current and overtaking lanes into two groups denoted by G^c and G^o, respectively, and order them along the road-aligned longitudinal position.
Exclude non-interacting objects which have higher speeds in front or lower speeds behind the vehicle platoon.
Control the vehicle platoon for LK if $G^c \neq \emptyset$.
Before overtaking:
  if $(G^o == \emptyset \parallel v_f^o == 0 \;\&\&\; (v_r^c == 0 \;\&\&\; d_c^o \geq d_s \parallel v_r^c > 0 \;\&\&\; s_f^p > s_f^o))) : M_o = 1$;
  elseif $(s_r^p \geq s_f^o + d_0) \;\&\&\; (v_r^p \geq v_f^o \parallel \tau == 0)$: $M_o = 4$;
  elseif $(v_r^o > v_r^c \;\&\&\; s_r^o \geq s_r^c)$: $M_o = 2$;
  else LK **End**.
  end if
During overtaking:
  if a new object arises in G^o and in front of the platoon before the step 3:
    if $((v_f^o == v_f^c == 0) \;\&\&\; (d_{c,f}^o \geq d_s))$: M_o: $2 \rightarrow 4$, otherwise remain M_o;
    else if $(v_f^o > v_f^c) : M_o$: $1 \rightarrow 3$, $4 \rightarrow 2$, otherwise remain M_o;
    else fail for overtaking, decelerate and then go back to the original lane **End**.
    end if
  end if
After overtaking: LK.

the lane changing maneuver. As a result, we construct the cost function q_p for the controlled platoon as follows:

$$q_p(x_t^p, u_t^p) = \lambda_s \frac{(\Delta v_t^{l,p})^2}{\Delta s_t^{l,p}} H(-\Delta v_t^{l,p}) + \lambda_r (v^r(\Delta s_t^{l,p}) - v_t^p)^2 + \lambda_c (\frac{\Delta v_t^{p,p}}{T_s})^2 + \lambda_e (\Delta v_t^{d,l})^2 H(\Delta v_t^{d,l}) \quad (14)$$

where

$$\Delta v_t^{l,p} = v_t^{l,\phi} - v_t^p, \quad \Delta s_t^{l,p} = s_t^{l,\phi} - s_t^p - l_v$$
$$\Delta v_t^{p,p} = v_t^p - v_{t-1}^p, \quad \Delta v_t^{d,l} = v^d - v_t^{l,\phi}$$

v^d is the desired velocity of the vehicle platoon preset by user and l_v is the length occupied by relevant vehicles. $v_t^{l,\phi}$ and $s_t^{l,\phi}$ denote the predecessor's velocity and position of the controlled platoon on lane ϕ, respectively:

$$\phi(\tau) = \begin{cases} original\ lane, & \text{if } t < \tau \\ overtaking\ lane, & \text{otherwise} \end{cases} \quad (15)$$

where τ is the time point to start a lane changing maneuver. To make the expression of (14) well-defined, $v_t^{l,\phi}$ and $s_t^{l,\phi}$ are set to $+\infty$ if there is no environment vehicle in front on lane ϕ at time point t. The local reference velocity of the vehicle platoon is denoted by v^r, which can be expressed as follow:

$$v^r(\Delta s_t^{l,p}) = \begin{cases} \frac{\Delta s_t^{l,p} - \Delta s_0}{t^g}, & \text{if } \Delta s_t^{l,p} < \Delta s^g \\ v^d, & \text{otherwise} \end{cases} \quad (16)$$

where Δs^0 is the minimum safety distance between vehicles and t^g is the given time gap. $\Delta s^g = v^d t^g + \Delta s_0$ represents the distance threshold below which the car-following mode is initiated. H is a unit step function which is defined as

$$H(x) = \begin{cases} 0, & \text{if } x < 0 \\ 1, & \text{otherwise} \end{cases} \quad (17)$$

Parameters λ_s, λ_r, λ_c and λ_e denote weighting factors. The first weighting term in (14) is related to safety, which inflicts a heavy penalty when the controlled platoon drives faster than its preceding vehicle with a small inter-vehicle distance. The second weighting term penalizes the deviation of the controlled platoon velocity from its local reference velocity v^r. The third weighting term reflects comfort by penalizing the rate of the control input. Traveling efficiency is indicated by the fourth weighting term, which incurs a penalty when the predecessor's velocity is lower than its desired velocity v^d.

The additional cost function q_r in (13) for the influenced environment vehicle j on the overtaking lane is given by

$$q_r(x_t^j, u_t^j) = \left(\lambda_s \frac{\left(\Delta v_t^{l,j}\right)^2}{\Delta s_t^{l,j}} H\left(-\Delta v_t^{l,j}\right) + \lambda_e \left(\Delta v_t^{d,p}\right)^2 H\left(\Delta v_t^{d,p}\right) \right) H(t-\tau) \quad (18)$$

where

$$\Delta v_t^{l,j} = v_t^l - v_t^j, \;\; \Delta s_t^{l,j} = s_t^l - s_t^j - l_v, \;\; \Delta v_t^{d,p} = v^{d,j} - v_t^p$$

Similarly, v_t^l and s_t^l represent the velocity and position of the preceding vehicle, respectively. For the first influenced vehicle, its predecessor is simply the controlled platoon. $v^{d,j}$ denotes the free driving velocity of the influenced environment vehicle j, i.e., the velocity before being intervened by the lane change actions of the platoon. The first cost term in (18) imposes a penalty for unsafe relative velocity and gap between vehicles. The second cost term penalizes undesired deceleration of environment vehicles due to the lane change maneuver. Equation (18) is constructed to guarantee that the overtaking process does not give rise to obvious decrease of traveling efficiency of environment vehicles.

Since it is difficult for the controlled platoon to obtain acceleration inputs of surrounding vehicles over a time horizon under a non-communication traffic environment, we use a Intelligent Driver Model (IDM) [37] to predict the accelerations responses of the influenced followers to the lane change action of the controlled platoon. Particularly, given an arbitrary input vector u^p of the controlled platoon, we can derive its longitudinal motion state x^p with the kinematic model, and then the control

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input and states of the first influenced rear adjacent vehicle can be estimated and predicted with the IDM and the same system model by using the known u^p and x^p as reference variables. The acceleration inputs of all other environment vehicles in N^r can be predicted in the same manner using their predecessor states as reference variables.

The objective of the optimization problem is to search for the optimal lane change timing and reference velocity sequence for the vehicle platoon over a prediction horizon T^o. This is formulated as the following finite time-horizon optimal control problem:

$$\min_{\tau\in\Pi, u^p_{1:N^o|t}} \sum_{k=0}^{N^o-1} q(s^p_{k+1|t}, v^p_{k+1|t}, \tau, s^j_{k+1|t}, v^j_{k+1|t}, a^j_{k|t}) \quad (19a)$$

$$\text{s.t. } x^p_{k+1|t} = A^p x^p_{k|t} + B^p u^p_{k|t} \quad (19b)$$

$$x^j_{k+1|t} = A^r x^j_{k|t} + B^r u^j_{k|t} \quad (19c)$$

$$u^j_{k|t} = \kappa(x^{j-1}_{k|t}, x^j_{k|t}, u^p_{k|t}) \quad (19d)$$

$$v_{\min} \le u^p_{k+1|t} \le v_{\max} \quad (19e)$$

$$\Delta v_{\min} \le u^p_{k+1|t} - u^p_{k|t} \le \Delta v_{\max} \quad (19f)$$

$$(j \in N^r,\ k = 0, \ldots, N^o - 1)$$

$$x_{0|t} = x_t,\ v^p_{0|t} = v^p_t \quad (19g)$$

where t denotes the current time point and $N^o = T^o/T_s$ is the maximum prediction step. $\Pi = \{0, T_s, \ldots, N^o T_s\}$ is the feasible set of the time points for lane change. (19b) and (19c) denote the kinematic models of the vehicle platoon and environment vehicles, respectively. (19d) denote the IDM. (19e) and (19f) enforce constraints on the magnitude and rate of the velocity. The initial conditions is given by (19g).

Solving the optimization problem (19) is a challenging task due to existence of lane change as a discrete variable. The computational complexity of the problem is closely related to the number of feasible solutions of lane change time point τ. Therefore, we first set the lane-change sampling frequency almost 10 times lower than sampling frequency of the vehicle steering input to reduce the feasible set Π of the solutions. Furthermore, we calculate the total predicted cost of the optimization problem over the finite time-horizon for each lane change time point τ. Then, the optimal lane changing time point among a finite number of feasible solutions is obtained by searching the minimum predicted cost.

Considering the different scenarios whether environment vehicles are influenced by the controlled platoon or not, a general decision-making logic for the overtaking maneuver is presented in Algorithm 1. The involved components on safety driving corridor constraints, target path planning and controller design are stated in Sections IV-C, respectively. In Algorithm 1, $v^\star_\dagger$ and $s^\star_\dagger$ denote the longitudinal velocity and position of the vehicle along the road centerline, respectively, where the subscript $\dagger \in \{f, r\}$ represents the vehicle order consistent with the first (f) and last (r) positions in a group of vehicles and the superscript $\star \in \{o, c, p\}$ is used for the group G^o (o), the group G^c (c), and the vehicle platoon (p), respectively, where the definition of G^o (o) and G^c (c) can be found in the following section. The design parameter d_l denotes the desired longitudinal displacement for lane change of the vehicle, which is defined and formulated in Section IV-E. M_o denotes the real-time overtaking mode. Parameter d_0 is the given longitudinal safety margin between vehicles. Values d^o_c and $d^o_{c,f}$ denote the relative distances from the group G^o to the group G^c and the first vehicle within G^c, respectively, which are defined as follows:

Algorithm 2: Iterative LTV-MPC Logic.

Initialization: $\bar{u}_{0:N-1|0} = 0$.
While $t \ge 0$ **Do**:
  $\xi_{0|t} = \xi_t$
  $\bar{u}_{0:N-1|t} = \text{RecedingFcn}(\bar{u}_{0:N-1|t-1})$
  $N_{it} = 0$
  While $\|u^\delta_{0:N-1|t}\| > \Phi_u$ && $N_{it} < N_{stop}$ **Do**:
$\bar{\xi}_{1:N|t} = \text{VehicleModel}(\xi_{0|t}, \bar{u}_{0:N-1|t})$
$u^\delta_{0:N-1|t} = \text{Solver}(\bar{u}_{0:N-1|t}, \bar{\xi}_{1:N|t})$
$\bar{u}_{0:N-1|t} = \bar{u}_{0:N-1|t} + u^\delta_{0:N-1|t}$
$N_{it} = N_{it} + 1$
  $u^*_t = \bar{u}_{0:|t}$
  $t = t + 1$

$$d^o_c = \begin{cases} s^o_r - s^c_f, & \text{if } s^o_r > s^c_f \\ s^c_r - s^o_f, & \text{if } s^c_r > s^o_f \\ 0, & \text{otherwise} \end{cases} \quad (20)$$

$$d^o_{c,f} = \begin{cases} s^o_r - s^c_f, & \text{if } M_o == 2 \\ s^o_f - s^c_f, & \text{otherwise.} \end{cases} \quad (21)$$

Where d_s represents the allowable minimum distance between two static obstacles occupying on different lanes, which guarantees a successful passing for the vehicle platoon at a safe speed. It is determined according to vehicle geometry.

Remark 2: To accomplish an overtaking maneuver, both speed and steering controls are required. As previously mentioned, we assume that longitudinal and lateral controls are decoupled and the former is realized by independent controllers. Nevertheless, the longitudinal and lateral controllers can share information, such as control inputs and determined overtaking mode, and operate in a collaborative manner. The longitudinal controllers perform as a CACC-based system that can regulate the vehicle platoon to track a given speed and a desired inter-vehicle spacing for the OA and LK maneuvers and to keep a safe distance from environment vehicle ahead.

C. Safe Driving Corridor

In this section, collision-free safety corridor constrains are formulated based on the environment perception model described in Section IV-A. The key idea is to match all objects within scope field $\mathcal{F}$ into a dynamic map along the road-aligned longitudinal coordinate s. As a result, the safety corridor constraints can simply be represented by the form of lower and upper boundaries without consideration of the road curvature. At an arbitrary prediction step k, the safety boundaries are calculated according

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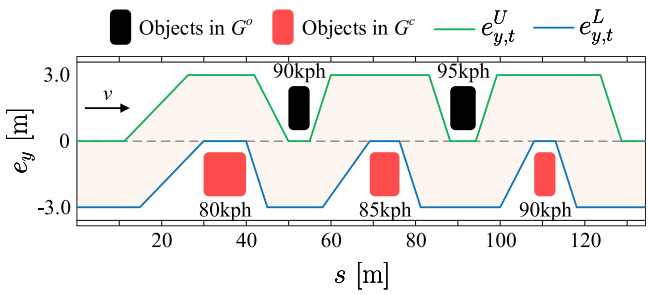

Fig. 7. Collision-free safety corridor constraints, where dashed blue and red lines denote the upper and lower boundaries, respectively.

to the predicted longitudinal positions and velocities of environment vehicles. Let $d_t^{r,1:N^c}$ ($v_t^{r,1:N^c}$) and $d_t^{r,1:N^o}$ ($v_t^{r,1:N^o}$) denote the relative longitudinal distances (velocities) from the controlled vehicles in the platoon to the environment vehicles in the groups G^c and G^o, respectively, where $1:N^c=\{1,\ldots,N^c\}$ and $1:N^o=\{1,\ldots,N^o\}$ are the vehicle sets of the groups G^c and G^o. For the simplicity of computation, we formulate the safety corridor constraints as time-varying piecewise line functions of $d_t^{r,1:N^c}$, $d_t^{r,1:N^o}$, $v_t^{r,1:N^c}$, and $v_t^{r,1:N^o}$, yielding

$$e^L_{y,t}\left(d_t^{r,1:N^c}, v_t^{r,1:N^c}\right) \le e_{y,t} \le e^U_{y,t}\left(d_t^{r,1:N^o}, v_t^{r,1:N^o}\right) \quad (22)$$

where $e^L_{y,t}$ and $e^U_{y,t}$ are respectively the lower and upper boundaries for the lateral displacement with respect to the centerline of the lane. The illustration is shown in Fig. 7. The slopes of the designed piecewise affine functions are related to the relative velocities $v_t^{r,1:N^c}$ ($v_t^{r,1:N^o}$). In our work, we linearly reduce the slopes with the increase of $v_t^{r,1:N^c}$ ($v_t^{r,1:N^o}$) based on the heuristic method in order to ensure enough longitudinal safety margins. The lower and upper boundaries $e^L_{y,t}$ and $e^U_{y,t}$ are dynamically updated based on the real-time measurement data $d_t^{r,1:N^c}$, $v_t^{r,1:N^c}$, $d_t^{r,1:N^o}$, and $v_t^{r,1:N^o}$, respectively. Note that Fig. 7 is only used for explanation, and the actual inter-vehicle distances in the groups G^c and G^o maybe large enough such that $e^L_{y,t}$ and $e^U_{y,t}$ are planned to pass among the vehicles of G^c and G^o, respectively.

Remark 3: A virtual lateral trajectory needs to be planned to provide the reference for the overtaking maneuver when the OA mode is activated. In this paper, the lateral reference for overtaking is generated based on a modified Cartesian polynomial curve [38].

D. Iterative Linear Time-Varying Model Predictive Control

MPC is a popular approach to develop the control strategy for autonomous driving due to its capabilities to deal with model nonlinearities and uncertainties, as well as system constraints in an integrated manner [39], [40]. In this section, with the vehicle dynamic model and safety driving corridor constraints, we formulate the lateral control design for an arbitrary vehicle i in the platoon using an LTV-MPC scheme. At every sampling time step, a constrained optimization problem is solved to yield a sequence of control input deviations by minimizing a predefined objective function. The first element of the sequence is then used for calculation of the final control input. At the subsequent sampling time step, the the optimization process is repeated after data updating over a forward-moving horizon, forming a receding horizon control strategy. The LTV-MPC problem for LK coupling with OA ca be depicted as follows:

$$\min_{u^{\delta,i}_{0:N-1|t}} \sum_{k=0}^{N-1} \|\bar{\eta}^i_{k+1|t} + \eta^{\delta,i}_{k+1|t} - \eta^{ref,i}_{k+1|t}\|^2_Q + \|\bar{u}^i_{k|t} + u^{\delta,i}_{k|t}\|^2_R$$
$$+ \|\Delta u^i_{k|t}\|^2_{R^\Delta} + \|u^{\delta,i}_{k|t}\|^2_{R^\delta} + \rho\epsilon \quad (23a)$$

$$\text{s.t.}\ \xi^{\delta,i}_{k+1|t} = A^i_{k|t}\xi^{\delta,i}_{k|t} + B^i_{k|t}u^{\delta,i}_{k|t} \quad (23b)$$

$$\eta^{\delta,i}_{k+1|t} = C\xi^{\delta,i}_{k+1|t} \quad (23c)$$

$$e^i_{y,k+1|t} \ge e^{L,i}_{y,k+1|t}(d^{r,1:N^c,i}_{k+1|t}, v^{r,1:N^c,i}_{k+1|t}) - \epsilon,$$
$$e^i_{y,k+1|t} \le e^{U,i}_{y,k+1|t}(d^{r,1:N^o,i}_{k+1|t}, v^{r,1:N^o,i}_{k+1|t}) + \epsilon \quad (23d)$$

$$\Delta u^i_{k|t} = (\bar{u}^i_{k|t} + u^{\delta,i}_{k|t}) - (\bar{u}^i_{k-1|t} + u^{\delta,i}_{k-1|t}) \quad (23e)$$

$$u_{\min} \le (\bar{u}^i_{k|t} + u^{\delta,i}_{k|t}) \le u_{\max} \quad (23f)$$

$$\dot{u}_{\min} \le \Delta u^i_{k|t}/T_s \le \dot{u}_{\max} \quad (23g)$$

$$\|u^{\delta,i}_{k|t}\| \le \epsilon_u \quad (23h)$$

$$(k=0,\ldots,N-1)$$

$$\epsilon \ge 0 \quad (23i)$$

$$\xi^{\delta,i}_{0|t} = \xi^i_t - \bar{\xi}^i_t, \quad u^{\delta,i}_{-1|t} = u^i_{t-1} - \bar{u}^i_{t-1} \quad (23j)$$

where the notation $\|x\|^2_M = x^T M x$, the current time point is indicated by the subscript t, and N denotes the maximum prediction step. Q, R, R^Δ, and R^δ denote tuning cost weights with appropriate dimensions. $\xi^{\delta,i}_{k|t}$ is the predicted state deviation at time $t+k$, which is computed by substituting the input deviation sequence $u^{\delta,i}_{0:N-1|t}$ into the linearized model (23b) based on the initial condition (23j). $\eta = [\,e_y, e_\psi\,]^T$ represents the output vector of the system, and the corresponding reference sequence $\eta^{ref,i}_{1:N|t}$ in the first additional term of (23a) produces the desired output performance (e.g. tracking the centerline of the road and avoiding obstacles). The second to the fourth augends in (23a) penalize the magnitude and rate of the input sequence as well as the deviation $u^{\delta,i}_{0:N-1|t}$ from the nominal values. The constraints related to the control inputs are introduced by (23e) to (23h). The recursive feasibility of the optimal control problem (23) is guaranteed by enforcing all the state constraints in (23d) as soft constraints with a slack variable ϵ. ϵ is penalized powerfully by a weight factor ρ in (23a) which is much greater than weight matrixes Q, R, R^Δ, and R^δ in order to keep ϵ at 0 in normal cases (i.e., keep the originally stipulated state constraints without the slack variable). When the optimal control problem (23) is infeasible, ϵ is increased to prioritize the feasibility. The input deviation constraint (23h) is employed to guarantee the precision of the linearized model about the nominal state sequence $\bar{\xi}^i_{1:N|t}$ and input sequence $\bar{u}^i_{0:N-1|t}$ which can be approximately calculated by

$$\bar{u}^i_{k-1|t} = \begin{cases} \bar{u}^i_{k|t-1}, & \text{if } k=1,2,\ldots,N-1 \\ \bar{u}^i_{N-1|t-1}, & \text{if } k=N. \end{cases} \quad (24)$$

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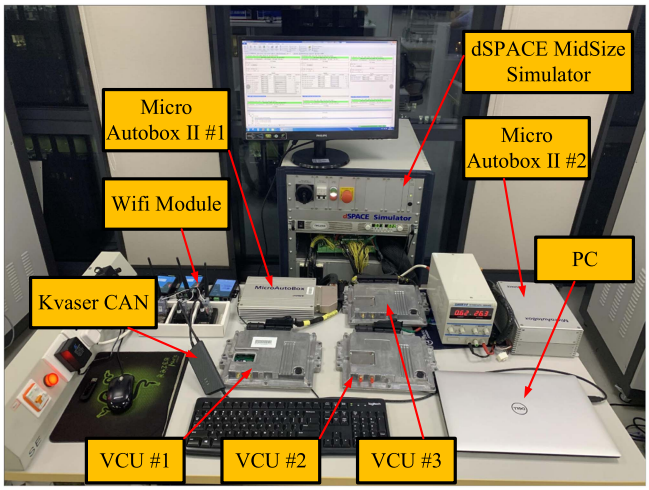

Fig. 8. HIL test platform.

The nominal output vector sequence $\bar{\eta}^i_{1:N|t}$ is obtained by applying $\bar{u}^i_{0:N-1|t}$ to the nonlinear dynamic model (10) in the form of discrete time by using the output (23c).

The solution of the optimization problem (23) at time instant t is denoted by $u^{\delta,i,\star}_{0:N-1|t}$. By using only the first element, the optimal control input at time t is given by the following feedback control law,

$$u^{i,\star}_t = \bar{u}^i_{0|t} + u^{\delta,i,\star}_{0|t}. \tag{25}$$

Since the vehicle model is locally linearized, the control accuracy cannot be guaranteed in some extreme situation (e.g., emergency obstacle avoidance) by the control laws (25). In this study, an ILTV-MPC method is proposed to ensure the feasibility and convergence of the LTV-MPC algorithm. The specific method is summarized in Algorithm 2, where the RecedingFcn() is (24), VehicleModel() is the linearized model in (23b) and (23c), Solver() represents the solver of the receding horizon problem and N_{stop} is the maximum iterative step. In addition, to guarantee the real-time performance of the ILTV-MPC algorithm, FORCES Pro is used to generate the solver code, which is highly optimized for fast computations and low memory footprint [41].

IV. Evaluation Results

In this section, experimental tests for the platooning formation consisting of one leading vehicle and two following vehicles are carried out to evaluate the performance of the proposed control strategy based on a real-time hardware-in-loop (HIL) simulation platform (see Fig. 8). The dSPACE-ASM-based vehicle dynamic models of leading vehicle and followers, which are customised from dSPACE GmbH, are compiled on the Midsize Simulator and two dSPACE MicroAutoBox II, respectively. In the test platform, the centralized decision-making algorithm is implemented on the PC, which performs as the computing platform on the leading vehicle. The distributed controller of each vehicle in the platoon is implemented on an embedded vehicle control unit (VCU) with a NVIDIA Jetson NANO module. The on-board communication is based on Kvaser CAN bus and the wireless communication between vehicles in platoon is based on kvaser CAN Wifi modules. The results of the simulation experiments and the driving states are observed by the ControlDesk interface in the monitor. To evaluate the performance, artificial potential field (APF) and platoon-lane-change (PLC) methods are used for comparison in experiments. The parameter sets of the proposed ILTV-MPC controller are listed in Table II.

TABLE II
Controller Parameters

Symbol	Description	Value [units]
N	Maximum prediction step	20 [-]
Q	Weight on reference trajectories	diag[10, 1] [-]
R	Weight on control input	0.1 [-]
R^{Δ}	Weight on control input rate	10 [-]
R^{δ}	Weight on control input deviation	15 [-]
ρ	Weight on constraint violation	10000 [-]
u_{min}, u_{max}	Constraints on control input	$\pm$20 [deg]
$\dot{u}_{min}, \dot{u}_{max}$	Constraints on control input rate	$\pm$10 [deg/s]
ϵ_u	Constraint on control input deviation	10 [deg]

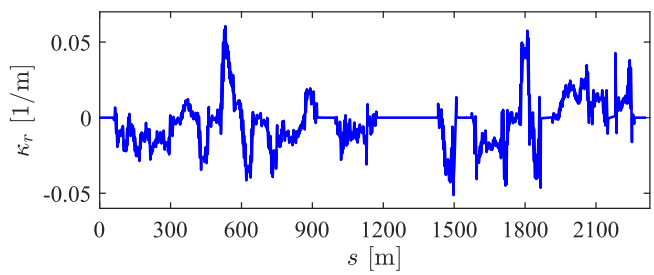

Fig. 9. Curvature of the test road.

A. Path Following

In order to evaluate tracking capabilities of the proposed ILTV-MPC controller, a path following experiment is performed on a designated two-lane test road. A non-iterative LTV-MPC controller is used for comparison. The test road consists of straights and curves with variable curvature radius. The goal is to track the lane centerline (i.e., setting $e^{ref}_y = -1.5$ for the right lane). The curvature information of the test road is displayed in Fig. 9. In this situation, we set the reference speed of the vehicle platoon to 90 kph for high speed experiment. The initial inter-vehicle distances within the platoon are configured to 10 m. Fig. 10 shows the tracking errors of leading vehicle controlled by two methods. It illustrates that the lateral tracking errors of iterative method is significantly less than those of the non-iterative method. In addition, Fig. 11 shows the values of control inputs of the proposed ILTV-MPC controller are all below the physical constraints.

B. Overtaking of Static Obstacles

Fig. 12 shows the case that the guided platoon passes two close static obstacles on different lanes. The positions of three platoon vehicles at some specific time points are displayed by small green, blue, and black squares, respectively. The red lines and

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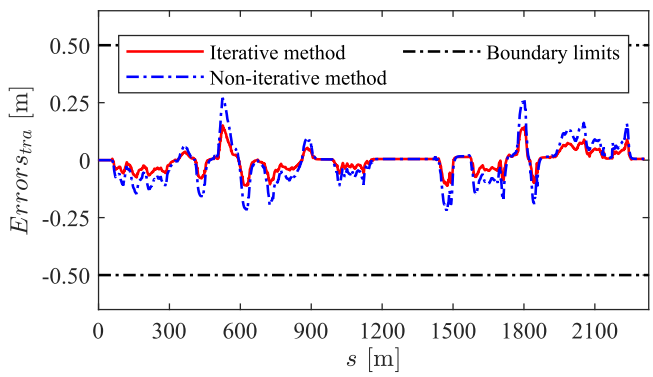

Fig. 10. Test results comparison of iterative and non-iterative methods.

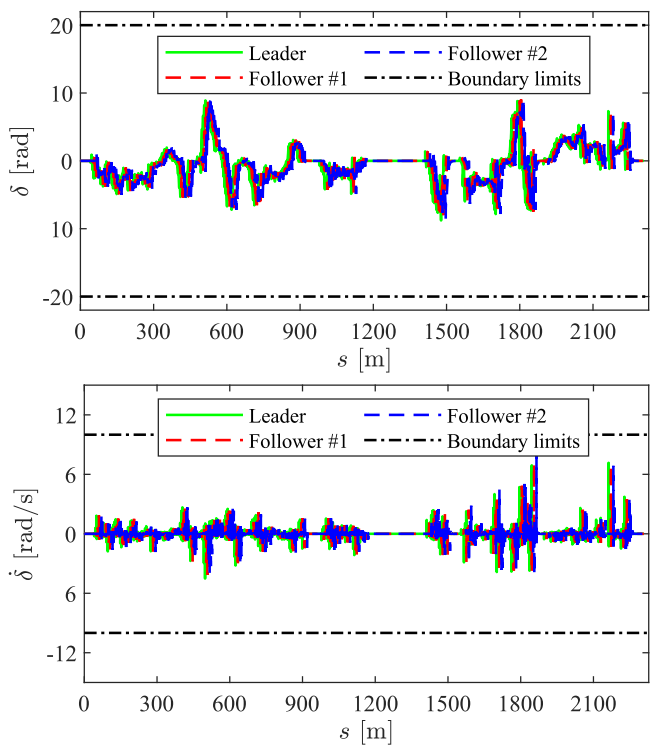

Fig. 11. Control inputs of ILTV-MPC controller.

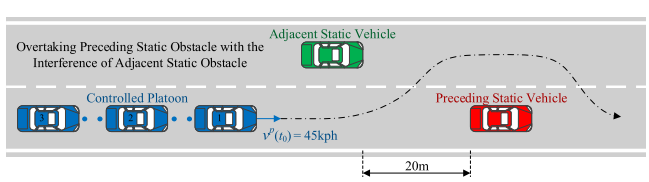

Fig. 12. Experiment scenarios of static obstacles at two lanes.

rectangles denote the road boundaries and obstacles (stopped vehicles), respectively. The numbers next to the vehicle platoon are specific time points. The initial inter-vehicle distance in the platoon is 10 m and the initial reference speed is 45 kph. $\mathcal{L}_f$ represents the view length to the front, which is 95 m. The performance of proposed method is shown in Fig. 13. As shown in the top and middle graphs, both the proposed method and APF can avoid two static vehicles without splitting. however, it is worth noting that the decision-making algorithm needs

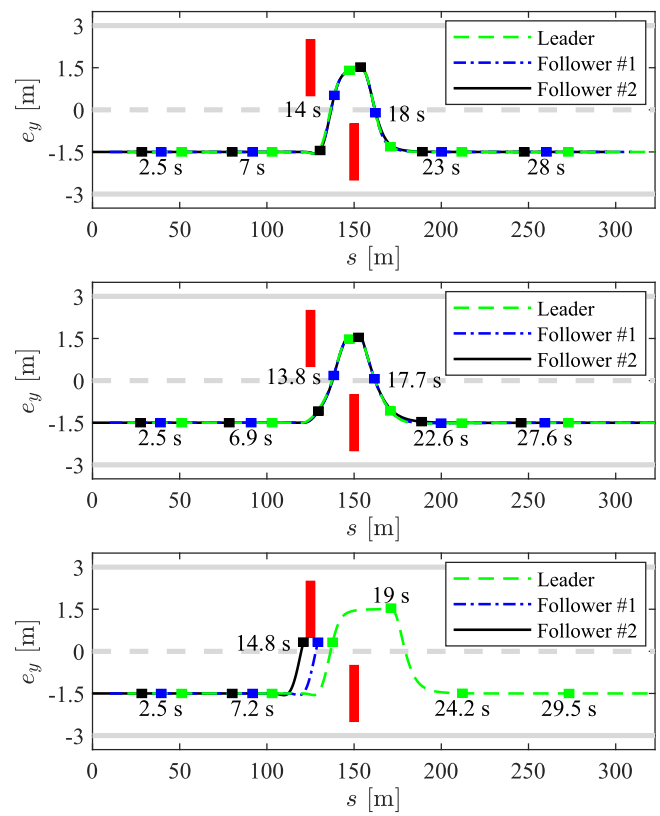

Fig. 13. Simulation results of two vehicles stopped on two different lanes of a straight road. Top graph: the proposed method. Middle graph: APF. Bottom graph: PLC.

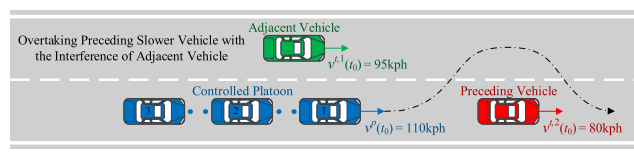

Fig. 14. The first experiment scenarios of moving obstacles.

to check if the distance between two static obstacles is larger than the minimum static overtaking distance d_s before passing. The performance of PLC is shown in the bottom graph. The platoon is divided into two parts, with two followers not finishing lane-change process.

C. Overtaking of Moving Obstacles

Two test scenarios involving two dynamic obstacles moving on different lanes are described.

In the first scenario (shown in Fig. 14), a slow vehicle (80 km/h) emerges in front when the vehicle platoon (initial velocity is 110 km/h) is tracking the centerline of the right lane and another slow vehicle (95 km/h) on the left lane is approaching from the front simultaneously. Figs. 15–18 show the simulation results of this scenario. The simulation results of proposed method are the same as those of PLC because the overtaking mode is mode 4 (two lane-changing patterns are both "uniformity"). The performances of proposed method and PLC are shown at the top graph in each figure and the performance of

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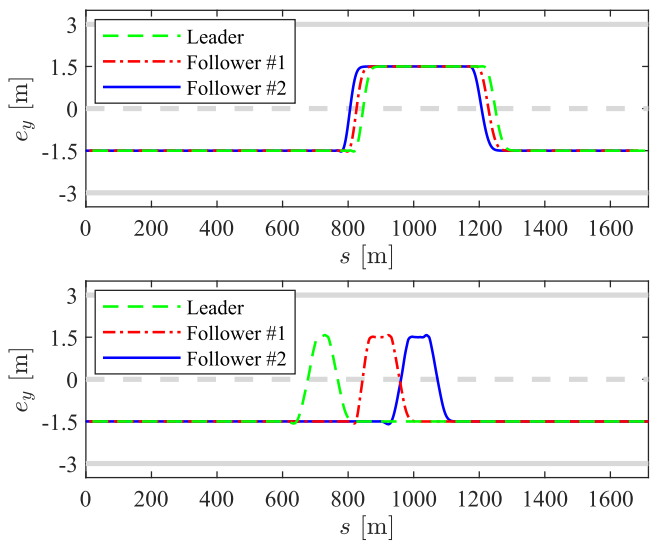

Fig. 15. Lateral offsets for the first scenario with moving obstacles. Top graph: the proposed method and PLC. Bottom graph: APF.

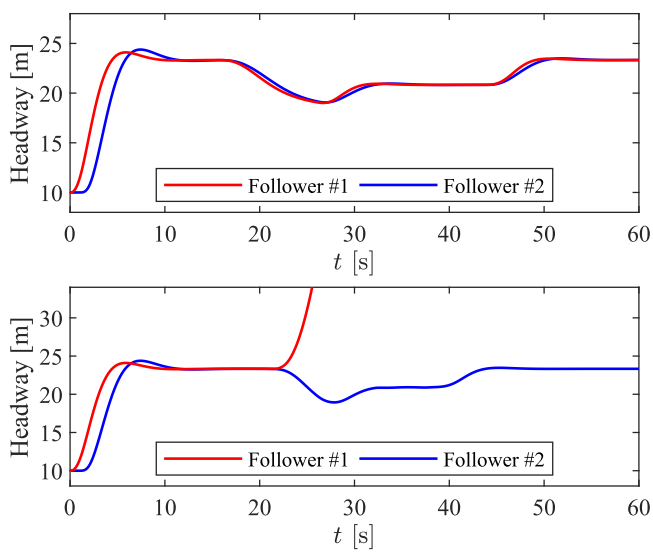

Fig. 16. Headway for the first scenario with moving obstacles. Top graph: the proposed method and PLC. Bottom graph: APF.

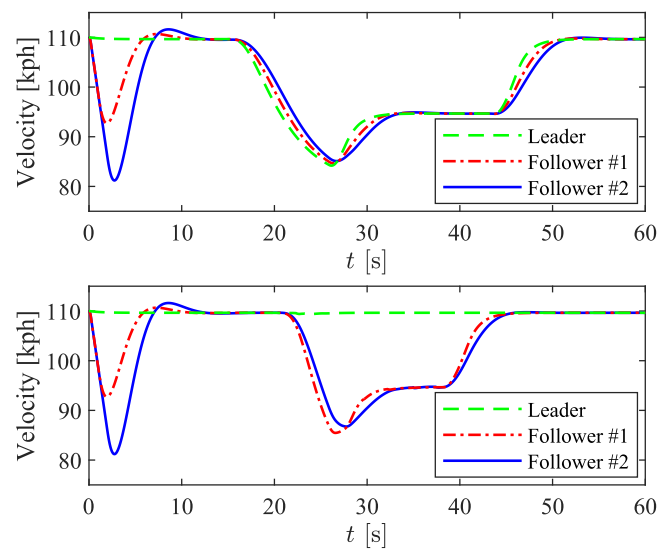

Fig. 17. Velocities for the first scenario with moving obstacles. Top graph: the proposed method and PLC. Bottom graph: APF.

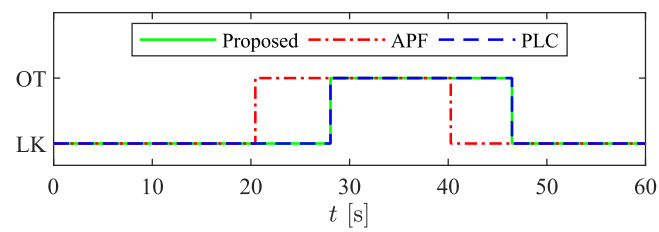

Fig. 18. Overtaking duration for the first scenario with moving obstacles.

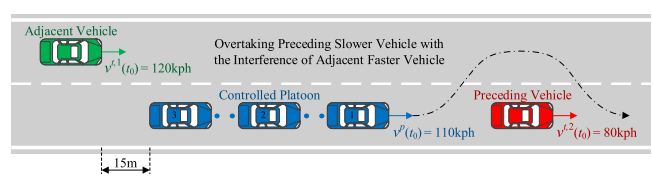

Fig. 19. The second experiment scenarios of moving obstacles.

APF is shown at the bottom graph. The lateral offsets in the first scenario are shown in Fig. 15 and the headways of followers are shown in Fig. 16. According to these two figures, we can see that the leading vehicle in the platoon finishes the OA process firstly by using APF, but the followers are split from the leading vehicle by the environment vehicle on the left lane, in contrast, the proposed method can keep the platoon intact.

The result can be further confirmed by the velocity of vehicles in different platoons in Fig. 17. Specifically, in the platoon using the proposed method, the decision-making algorithm is activated firstly, which controls the entire platoon to wait for proper time before overtaking, when the environment vehicle on left lane catches up with another one on right lane, the entire platoon performs the overtaking maneuver while keeping intact. In the platoon using APF, the leading vehicle keeps the initial velocity (110 km/h) and performs the overtaking maneuver without braking and waiting for the proper time for overtaking, thus the platoon is split. The overtaking duration is shown in Fig. 18. It indicates that the platoon with APF finishes the overtaking maneuver faster, but the overtaking duration for the platoon using the proposed method is shorter.

In the second scenario (shown in Fig. 19), a slow vehicle (80 km/h) emerges in front of the vehicle platoon, and a faster vehicle (120 km/h) on the left lane is approaching from behind. The corresponding simulation results can be seen in Figs. 20–23. The performances of proposed method are shown at the top graph in each figure while those of APF and PLC are shown at the middle and bottom graphs, respectively. The lateral offsets are shown in Fig. 20 and the headways of followers are shown in Fig. 21. According to the two figures, we can see that the leading vehicle using proposed method and APF finish the OA process

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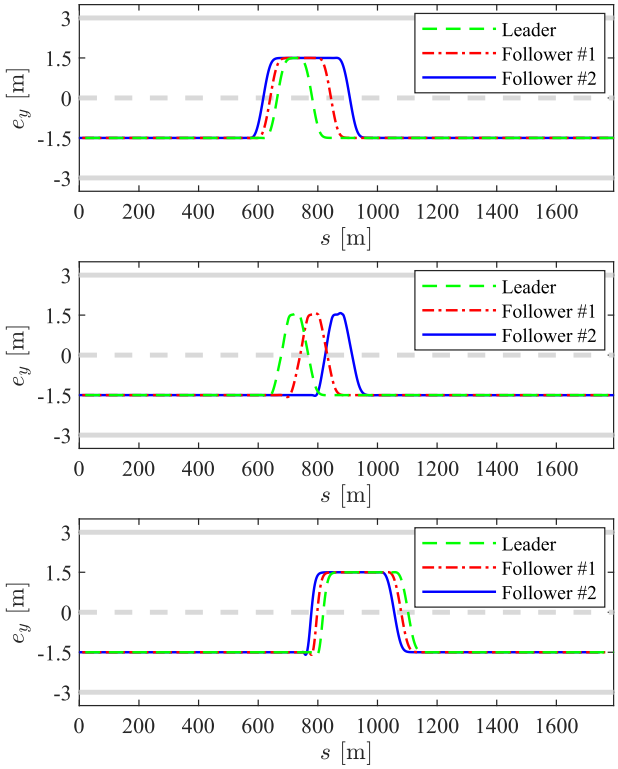

Fig. 20. Lateral offsets for the second scenario with moving obstacles. Top graph: the proposed method. Middle graph: APF. Bottom graph: PLC.

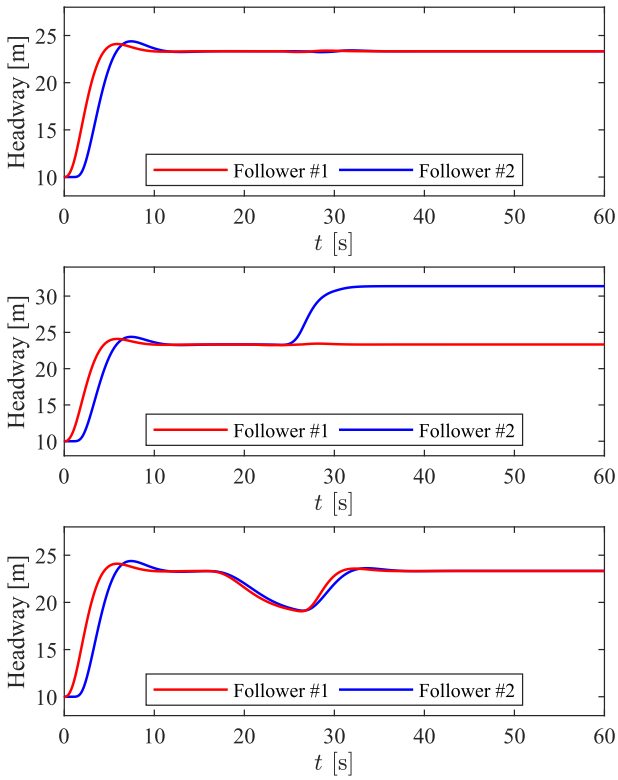

Fig. 21. Headway for the second scenario with moving obstacles. Top graph: the proposed method. Middle graph: APF. Bottom graph: PLC.

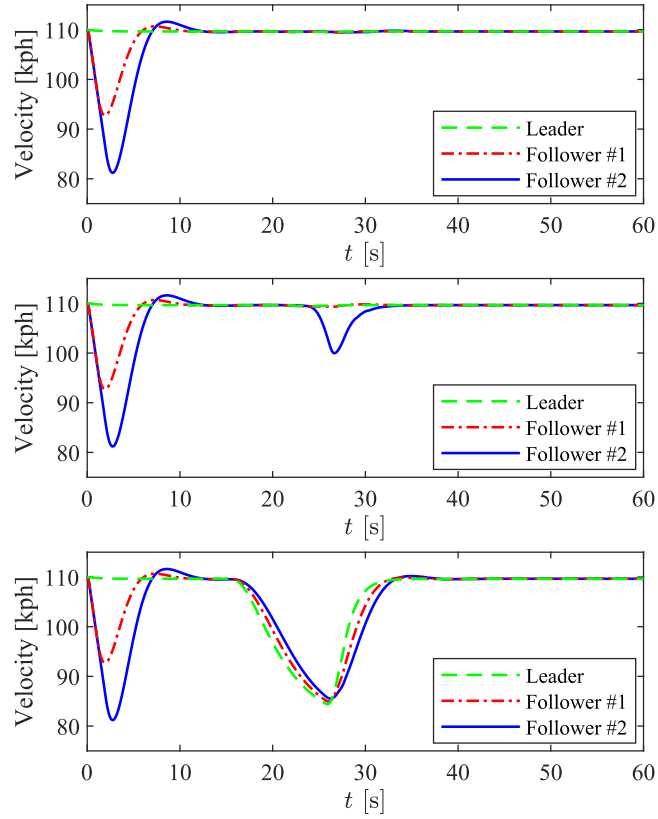

Fig. 22. Velocities for the second scenario with moving obstacles. Top graph: the proposed method. Middle graph: APF. Bottom graph: PLC.

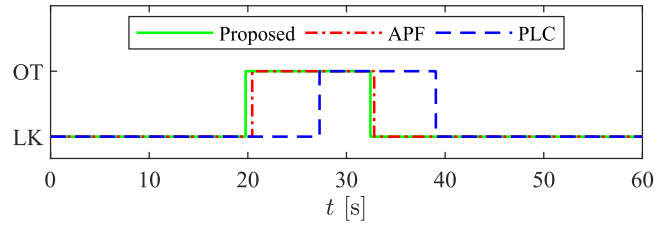

Fig. 23. Overtaking duration for the second scenario with moving obstacles.

faster than that using PLC, but the last vehicle in platoon using APF are split from the other vehicles by the fast environment vehicle on the left lane. Both the proposed method and PLC can keep the platoon intact during OA process.

These results are further confirmed by the velocity of vehicles in different platoons in Fig. 22. In the proposed method, the decision-making algorithm makes the entire platoon perform the first lane-changing maneuver in "uniformity" mode and perform the second lane-changing maneuver in "snake" mode, where the platoon keeps initial velocity in both two patterns. However, in the APF method, the leading vehicle and first follower keep the initial velocity and perform the overtaking maneuver successfully, but the last follower is separated from the platoon. In addition, it can be seen from the bottom graph that the PLC method chooses a conservative way and waits for the fast environment vehicle passing. The overtaking duration shown in Fig. 23 indicates that only the platoon with the proposed method

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TABLE III
THE COST TERMS UNDER DIFFERENT OVERTAKING TIME INSTANTS

Overtaking time	Traveling cost	Additional cost	Total cost
$\tau = 0$	56	36	92
$\tau = 1$	62	53	115
$\tau = 2$	75	137	212
$\tau = 3$	85	579	664
$\tau = 4$	94	1459	1553
$\tau = 5$	98	10^{10}	10^{10}
$\tau = 6$	100	10^{11}	10^{11}
$\tau = 7$	104	10^{10}	10^{10}
$\tau = 8$	110	0	110
$\tau = 9$	114	0	114
No overtaking	128	0	128

can perform the OA maneuver fast while keeping the platoon intact. In this case, the driving mode of the environment vehicle in adjacent lane can be influenced by the overtaking maneuver of controlled platoon, thus the game theoretic decision-making algorithm is activated and the interactive cost function is calculated at each overtaking time point, as shown in Table III. The total cost results indicate that the optimal overtaking time point is at $\tau = 0$.

In summary, the proposed control strategy can handle the different overtaking scenarios involving multiple moving obstacles, especially when the driving modes of environment vehicles may be influenced by the controlled platoon.

D. General Case

The experimental test involving multiple static and dynamic obstacles is performed on a two-lane curved road. Snapshots of simulation at different time points are shown in Fig. 24. The lateral offsets along the longitudinal position of the road and overtaking mode of the vehicle platoon are illustrated in Fig. 25. Since we describe the lateral position error model of the vehicle along the road centerline, the lateral offsets on a curved road can still be represented with the form of the upper and lower boundaries, which is consistent with that on the straight road. The initial reference speed of the vehicle platoon in this test is 45 kph.

In Fig. 24, the gray area denotes the view field $\mathcal{F}$. The light blue area represents the area where the overtaking of vehicle platoon is prohibited as the adjacent lane is occupied by environment vehicles. The entire process can be divided into seven stages as follows: (a) The overtaking mode 1 is activated to pass two stopped vehicles at a speed of 45 kph. (b) Two vehicles whose speeds are respectively 0 and 30 kph on the left and right lanes, respectively, emerge in front while the vehicle platoon is passing the second stopped vehicle. (c) A vehicle (37.9 kph) on the left lane (a magenta rectangle) is suddenly detected when the vehicle platoon is passing the front moving vehicle with the overtaking mode 4. (d) The vehicle platoon adjusts its overtaking mode to 2 and regulates its speed to follow the vehicle on the left lane until the vehicle on the right lane is passed. (e) The platoon decelerates again to follow the front vehicle (33 kph) as the left lane is blocked (i.e., ACC). (f) The platoon has passed

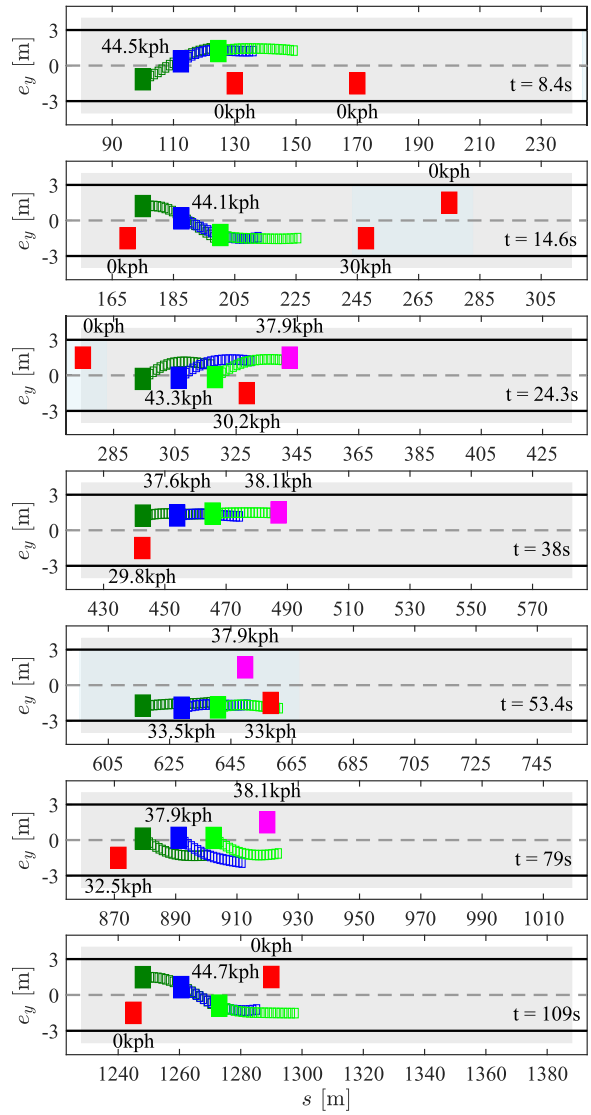

Fig. 24. Snapshots of simulation for a general case on a curved road.

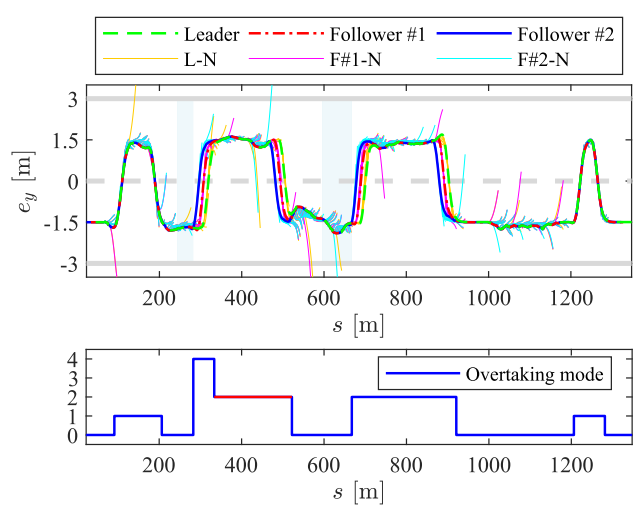

Fig. 25. simulation results for a general case on a curved road. Top graph: the lateral offsets of the platoon. Bottom graph: overtaking modes.

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TABLE IV
THE AVERAGE/MAXIMUM COMPUTING TIME OF THE PROPOSED ALGORITHMS

Algorithm	average computing time [ms]	maximum computing time [ms]
Centralized decision-making	184	233
ILTV-MPC lateral control	47	82

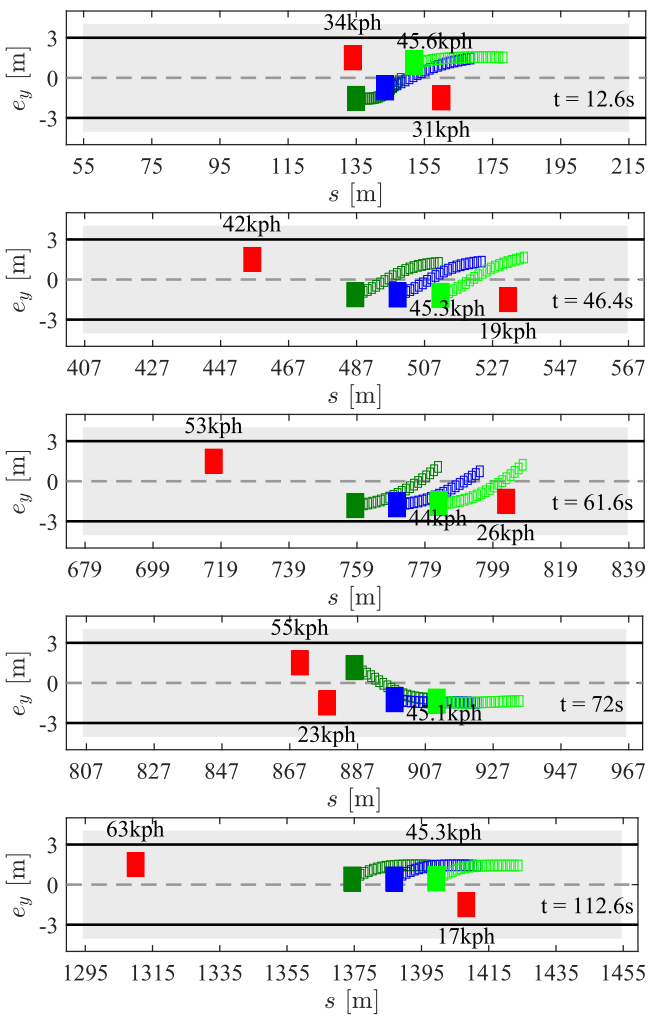

Fig. 26. Snapshots of the first case with random environment vehicles.

the front moving vehicle with the overtaking mode 2 and is returning to the right lane. (g) The platoon is overtaking two static vehicles with the overtaking mode 1. It can be seen that the guided vehicle platoon finishes the general case simulation without splitting by performing the overtaking maneuvers for four times and reducing its speed to follow the front moving obstacles for three times.

In Fig. 25, the actual paths of the vehicles in the platoon are represented by Leader, Follower #1, and Follow #2, respectively. The corresponding N-step predicted paths solved by the MPC controllers at every updating step are denoted by L-N, F#1-N, and F#2-N, respectively. The top graph shows the lateral offsets of the vehicle platoon and the bottom graph shows overtaking modes in different stages. In addition to four overtaking modes defined in Section IV-B, $\{1, 2, 3, 4\}$, the LK mode is added,

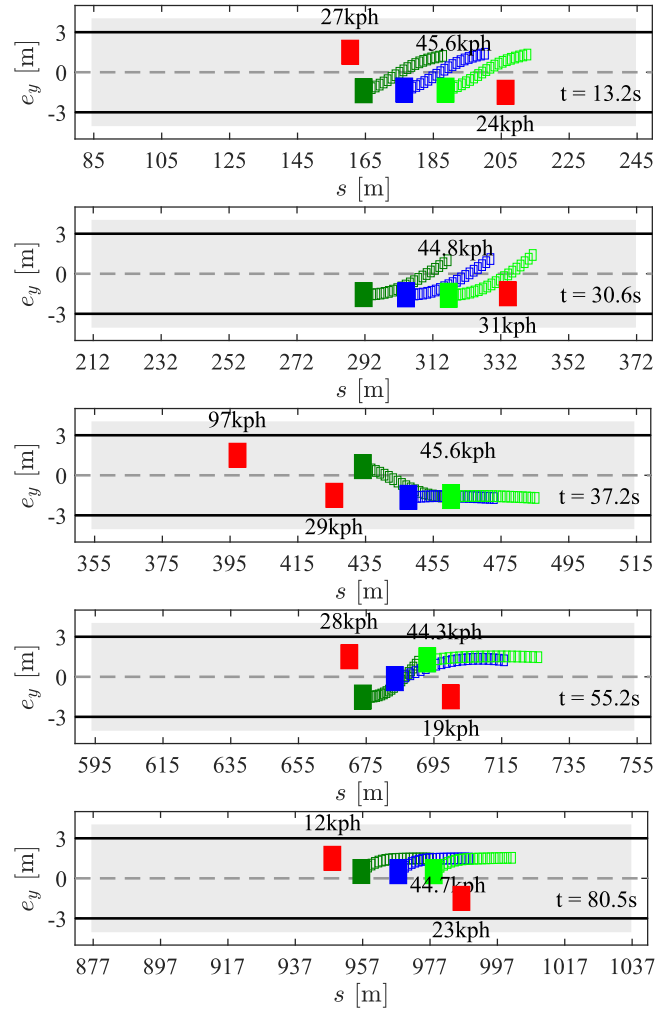

Fig. 27. Snapshots of the second case with random environment vehicles.

which is denoted by 0. It indicates that the overtaking mode can be switched in case of a sudden change in environment vehicles in the view field $\mathcal{F}$. Particularly, the red line in the bottom graph represents that the overtaking mode is adjusted from 4 to 2 due to the detected magenta rectangle obstacle during the second overtaking process.

In addition, to evaluate the real-time performance, the average/maximum computing time of each component in the proposed framework is shown in Table IV. It illustrates that both the centralized decision-making algorithm and the ILTV-MPC lateral controller can finish the calculation within a sampling time (i.e., 1 s and 100 ms, respectively).

E. Cases With Random Environment Vehicles

To evaluate the performance of the proposed method in random traffic flow, two cases with random environment vehicles are designed. Both of the position and initial speed of each vehicle are set randomly. The snapshots of overtaking maneuvers in two cases are shown in Figs. 26 and 27. It can be seen that the guided vehicle platoon can select the proper overtaking mode in different conditions to avoid splitting in both two cases, and

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the speed of the platoon also can be controlled around the target speed (45 km/h). The results indicate that the proposed method can keep its effectiveness in random traffic flow.

In conclusion, the proposed decision-making and lateral control framework can select the proper overtaking mode and perform the overtaking maneuver safely and effectively with varying scenarios.

V. Conclusion

In this paper, an integrated decision-making and lateral control approach for guided platooning of autonomous vehicles is presented. The main capability of the proposed approach is cooperative obstacle avoidance (OA) for the guided vehicle platoon in varying scenarios. In the control framework, a global decision-making approach is developed to cope with dynamic OA scenarios. In addition, a distributed iterative linear time-varying model predictive control (ILTV-MPC) method combined with a safe driving corridor planner is proposed for lateral motion control. The performance of the proposed method is evaluated by experiments involving static and dynamic obstacles based on the dSPACE HIL test platform. The results show that the proposed game theory-based decision-making algorithm and the ILTV-MPC lateral control method can (1) track the reference path accurately, (2) perform overtaking maneuvers in the scenarios involving multiple static and dynamic obstacles without splitting, (3) implement efficient and safe cooperative OA under varying scenarios with unexpected situations and randomness.

Future work will integrate longitudinal control into the system architecture and evaluate the proposed method in more complex traffic flow. Moreover, the uncertainties of the vehicle models as well as the estimation of driving intentions of surrounding vehicles need to be systematically considered in subsequent researches.

References

[1] S. B. Prathiba, G. Raja, K. Dev, N. Kumar, and M. Guizani, "A hybrid deep reinforcement learning for autonomous vehicles smart-platooning," *IEEE Trans. Veh. Technol.*, vol. 70, no. 12, pp. 13340–13350, Dec. 2021.

[2] Z. Yang, J. Huang, D. Yang, and Z. Zhong, "Collision-free ecological cooperative robust control for uncertain vehicular platoons with communication delay," *IEEE Trans. Veh. Technol.*, vol. 70, no. 3, pp. 2153–2166, Mar. 2021.

[3] S. Feng, Z. Song, Z. Li, Y. Zhang, and L. Li, "Robust platoon control in mixed traffic flow based on tube model predictive control," *IEEE Trans. Intell. Transp. Syst.*, vol. 6, no. 4, pp. 711–722, Dec. 2021.

[4] S. E. Li, Z. Jia, K. Li, and B. Cheng, "Fast online computation of a model predictive controller and its application to fuel economy–oriented adaptive cruise control," *IEEE Trans. Intell. Transp. Syst.*, vol. 16, no. 3, pp. 1199–1209, Sep. 2015.

[5] F. Ma et al., "Distributed control of cooperative vehicular platoon with nonideal communication condition," *IEEE Trans. Veh. Technol.*, vol. 69, no. 8, pp. 8207–8220, Aug. 2020.

[6] Y. Zhu, D. Zhao, and H. He, "Synthesis of cooperative adaptive cruise control with feedforward strategies," *IEEE Trans. Intell. Transp. Syst.*, vol. 69, no. 4, pp. 3615–3627, Apr. 2020.

[7] R. White and M. Tomizuka, "Autonomous following lateral control of heavy vehicles using laser scanning radar," in *Proc. IEEE Amer. Control Conf.*, Arlington, VA, USA, vol. 3, 2001, pp. 2333–2338.

[8] M. Liu, K. Chour, S. Rathinam, and S. Darbha, "Lateral control of an autonomous and connected following vehicle with limited preview information," *IEEE Trans. Intell. Transp. Syst.*, vol. 6, no. 3, pp. 406–418, Sep. 2021.

[9] J. A. Marshall, M. E. Broucke, and B. A. Francis, "Formations of vehicles in cyclic pursuit," *IEEE Trans. Robot. Autom.*, vol. 49, no. 11, pp. 1963–1974, Nov. 2004.

[10] D. M. Stipanović, G. Inalhan, R. Teo, and C. J. Tomlin, "Decentralized overlapping control of a formation of unmanned aerial vehicles," *Automatica*, vol. 40, no. 8, pp. 1285–1296, Aug. 2004.

[11] H. G. Tanner, G. J. Pappas, and V. Kumar, "Leader-to-formation stability," *IEEE Trans. Robot. Autom.*, vol. 20, no. 3, pp. 443–455, Jun. 2004.

[12] I. Papadimitriou and M. Tomizuka, "Lateral control of platoons of vehicles on highways: The autonomous following based approach," *Int. J. Veh. Des.*, vol. 36, no. 1, pp. 24–37, Jan. 2004.

[13] L. Xu, W. Zhuang, G. Yin, C. Bian, and H. Wu, "Modeling and robust control of heterogeneous vehicle platoons on curved roads subject to disturbances and delays," *IEEE Trans. Veh. Technol.*, vol. 68, no. 12, pp. 11551–11564, Dec. 2019.

[14] R. Firoozi, X. Zhang, and F. Borrelli, "Formation and reconfiguration of tight multi-lane platoons," *Control Eng. Pract.*, vol. 108, 2021, Art. no. 104714.

[15] L. Gao, D. Chu, Y. Cao, L. Lu, and C. Wu, "Multi-lane convoy control for autonomous vehicles based on distributed graph and potential field," in *Proc. IEEE Intell. Transp. Syst. Conf.*, 2019, pp. 2463–2469.

[16] H. Peng and M. Tomizuka, "Vehicle lateral control for highway automation," in *Proc. Amer. Control Conf.*, San Diego, USA, 1990, pp. 788–794.

[17] S. E. Shladover et al., "Automated vehicle control developments in the PATH program," *IEEE Trans. Veh. Technol.*, vol. 40, no. 1, pp. 114–130, Feb. 1991.

[18] P. Hingwe and M. Tomizuka, "Experimental evaluation of a chatter free sliding mode control for lateral control in AHS," in *Proc. IEEE Amer. Control Conf.*, Albuquerque, NM, USA, vol. 5, 1997, pp. 3365–3369.

[19] R. Rajamani, H.-S. Tan, B. K. Law, and W.-B. Zhang, "Demonstration of integrated longitudinal and lateral control for the operation of automated vehicles in platoons," *IEEE Trans. Control Syst. Technol.*, vol. 8, no. 4, pp. 695–708, Jul. 2000.

[20] S. Tsugawa, S. Kato, and K. Aoki, "An automated truck platoon for energy saving," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, San Francisco, CA, USA, 2011, pp. 4109–4114.

[21] G. C. Keler, J. P. Maschuw, A. Bollig, and D. Abel, "Lateral guidance of heavy-duty vehicle platoons using model-based predictive control," *IFAC Proc. Volumes*, vol. 39, no. 12, pp. 433–438, Jan. 2006.

[22] O. McAree and S. M. Veres, "Lateral control of vehicle platoons with on-board sensing and inter-vehicle communication," in *Proc. Eur. Control Conf.*, Aalborg, Denmark, 2016, pp. 2465–2470.

[23] R. Kianfar, P. Falcone, and J. Fredriksson, "A distributed model predictive control approach to active steering control of string stable cooperative vehicle platoon," *IFAC Proc. Volumes*, vol. 46, no. 21, pp. 750–755, Jan. 2013.

[24] S. Solyom, A. Idelchi, and B. B. Salamah, "Lateral control of vehicle platoons," in *Proc. IEEE Int. Conf. Syst. Man, Cybern.*, Manchester, U.K., 2013, pp. 4561–4565.

[25] X. Huang, H. Zhang, G. Zhang, and J. Wang, "Robust weighted gain-scheduling H_∞ vehicle lateral motion control with considerations of steering system backlash-type hysteresis," *IEEE Trans. Control Syst. Technol.*, vol. 22, no. 5, pp. 1740–1753, Sep. 2014.

[26] R. Benton and D. Smith, "A static-output-feedback design procedure for robust emergency lateral control of a highway vehicle," *IEEE Trans. Control Syst. Technol.*, vol. 13, no. 4, pp. 618–623, Jul. 2005.

[27] H. Du, N. Zhang, and G. Dong, "Stabilizing vehicle lateral dynamics with considerations of parameter uncertainties and control saturation through robust yaw control," *IEEE Trans. Veh. Technol.*, vol. 59, no. 5, pp. 2593–2597, Jun. 2010.

[28] S. Mata, A. Zubizarreta, and C. Pinto, "Robust tube-based model predictive control for lateral path tracking," *IEEE Trans. Intell. Veh.*, vol. 4, no. 4, pp. 569–577, Dec. 2019.

[29] C. Lin, X. Gong, R. Xiong, and X. Cheng, "A novel H_∞ and EKF joint estimation method for determining the center of gravity position of electric vehicles," *Appl. Energy*, vol. 194, pp. 609–616, May 2017.

[30] X. Gong, J. Suh, and C. Lin, "A novel method for identifying inertial parameters of electric vehicles based on the dual H infinity filter," *Veh. Syst. Dyn.*, vol. 58, no. 1, pp. 28–48, 2020. [Online]. Available: https://doi.org/10.1080/00423114.2019.1566559

[31] Y. Guan et al., "Learn collision-free self-driving skills at urban intersections with model-based reinforcement learning," in *Proc. IEEE Int. Intell. Transp. Syst. Conf.*, 2021, pp. 3462–3469.

[32] Y. Guan et al., "Integrated decision and control: Toward interpretable and computationally efficient driving intelligence," *IEEE Trans. Cybern.*, vol. 53, no. 2, pp. 859–873, Feb. 2023.

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[33] J. Jiang et al., "Integrated decision and control at multi-lane intersections with mixed traffic flow," *J. Phys. Conf.*, vol. 2234, no. 1, pp. 012–015, 2022.
[34] P. Petrov and F. Nashashibi, "Modeling and nonlinear adaptive control for autonomous vehicle overtaking," *IEEE Trans. Intell. Transp. Syst.*, vol. 15, no. 4, pp. 1643–1656, Aug. 2014.
[35] J. Ji, A. Khajepour, W. W. Melek, and Y. Huang, "Path planning and tracking for vehicle collision avoidance based on model predictive control with multiconstraints," *IEEE Trans. Veh. Technol.*, vol. 66, no. 2, pp. 952–964, Feb. 2016.
[36] M. Werling, J. Ziegler, S. Kammel, and S. Thrun, "Optimal trajectory generation for dynamic street scenarios in a Frenet Frame," in *Proc. IEEE Int. Conf. Robot. Autom.*, Anchorage, AK, USA, 2010, pp. 987–993.
[37] M. Treiber, A. Hennecke, and D. Helbing, "Congested traffic states in empirical observations and microscopic simulations," *Phys. Rev. E*, vol. 62, no. 2, pp. 1805–1824, Aug. 2000.
[38] W. Nelson, "Continuous-curvature paths for autonomous vehicles," in *Proc. IEEE Int. Conf. Robot. Autom.*, Scottsdale, AZ, USA, 1989, pp. 1260–1264.
[39] P. Falcone, F. Borrelli, J. Asgari, H. E. Tseng, and D. Hrovat, "Predictive active steering control for autonomous vehicle systems," *IEEE Trans. Control Syst. Technol.*, vol. 15, no. 3, pp. 566–580, May 2007.
[40] F. Borrelli, P. Falcone, T. Keviczky, J. Asgari, and D. Hrovat, "MPC-based approach to active steering for autonomous vehicle systems," *Int. J. Veh. Auton. Syst.*, vol. 3, no. 2, pp. 265–291, Jan. 2005.
[41] A. Domahidi and J. Jerez, "Forces Professional," embotech GmbH, Jul. 2014. [Online]. Available: http://embotech.com/FORCES-Pro

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